

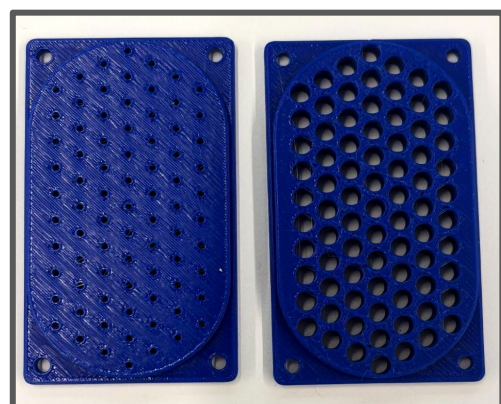
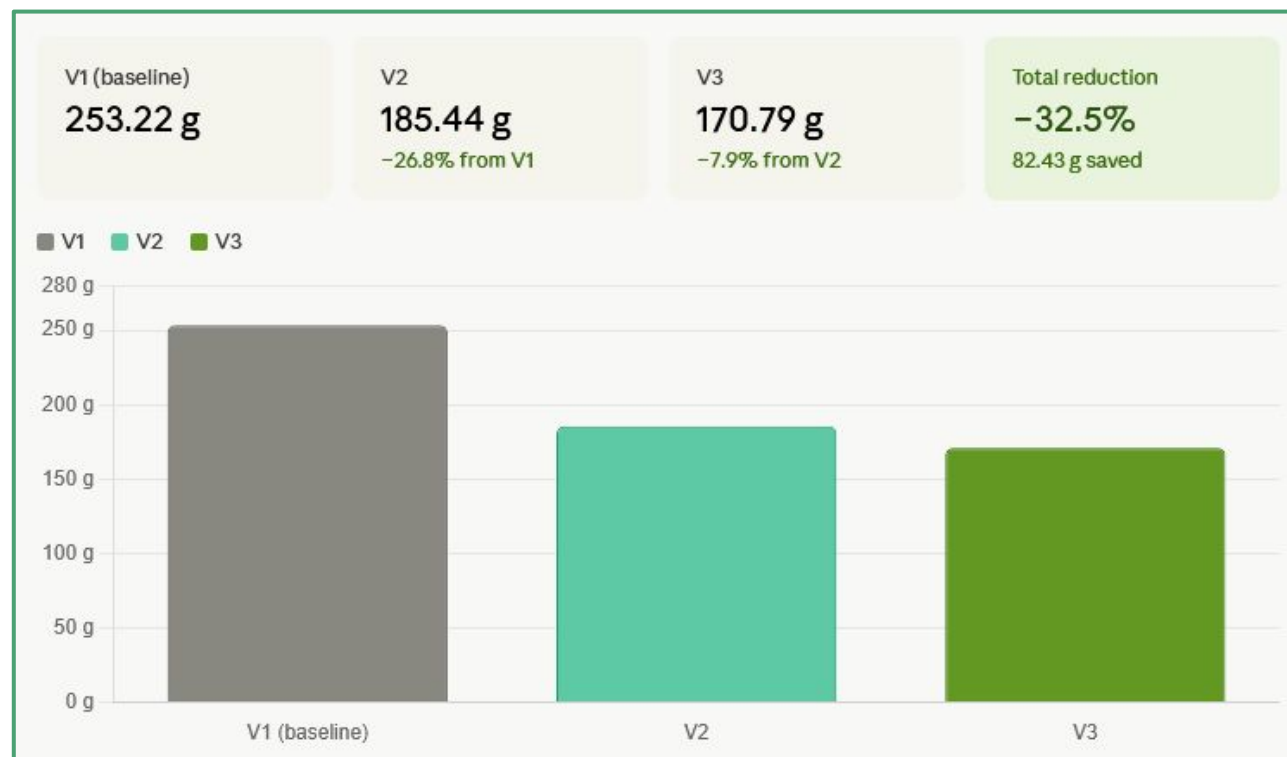
Opportunity

Creating a customizable device that takes physical input and turns it into audio output to improve the communication of non-verbal children.

Key Design Specifications:

- Sound level, Decibels (dB), ↑
- Battery Life, Hours, ↑
- Portability, Grams(g), ↓

Main Device Material Reduction



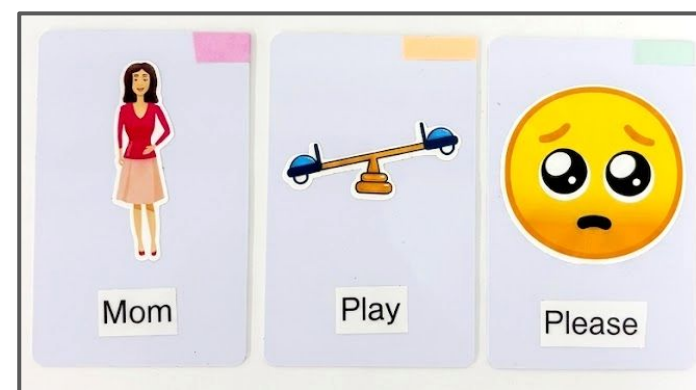
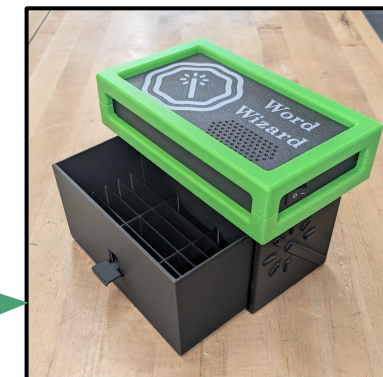
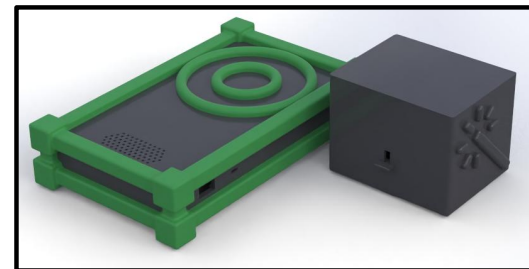
Tested Speaker Grill:

- Tested 6 diameters, + control
- 2.5mm had loudest output
- Smaller holes muffled output
- Larger holes prone to debris

Word Wizard



Changes to Design:



NFC Cards:

- Each card is programmable
- Split into groups by word type
 - Colored tape on corner
- 30 different cards included

Broader Impacts:

- Has high potential as a screen free learning tool, allowing for a larger market

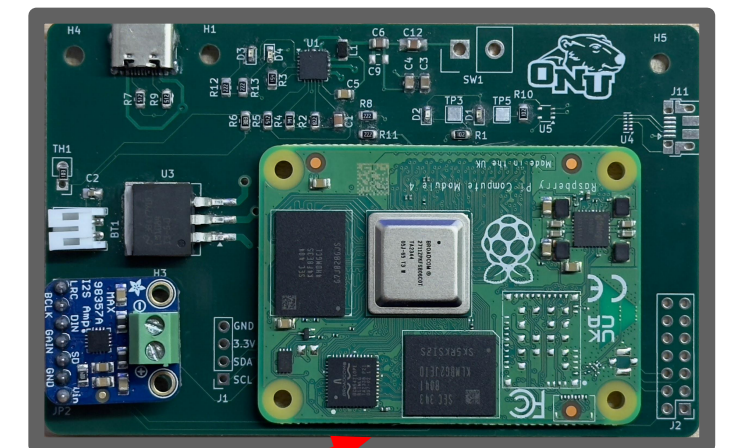
Key Research:

- There is a need for a mid tech non-verbal communication solution
- Decibel level of 60-65 dB is needed for high quality sound output

Next Steps:

- Create a simplified storage system for easier access.
- Creating a single device by merging storage with the main device.

Assembled PCB



Raspberry Pi